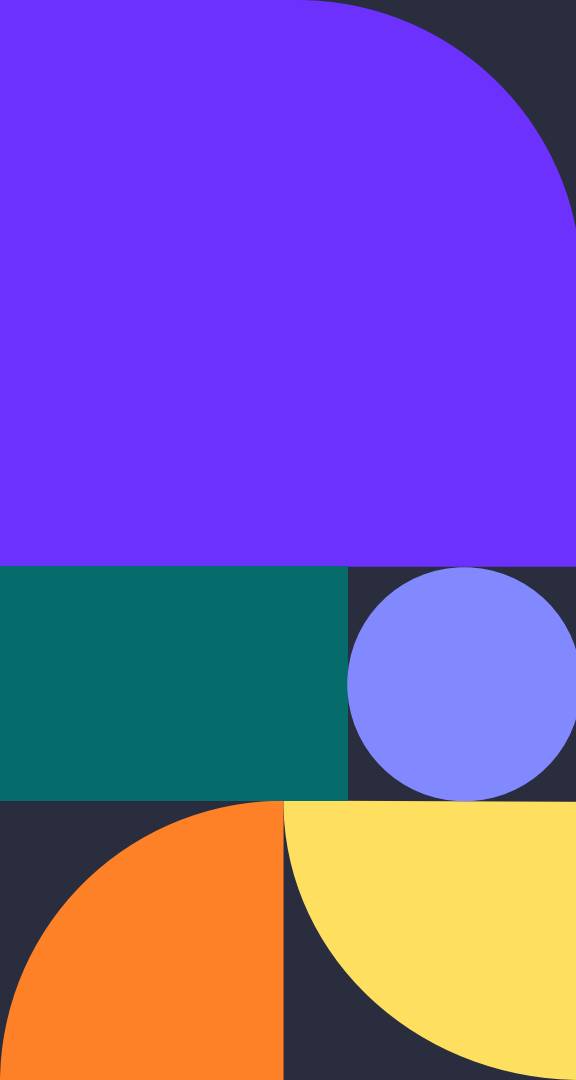




Diabetes Transition Navigator: A Cost-Sensitive Decision-Support System for Reducing 30-Day Readmissions

By: Muskaan Mahes

Business Problem and Stakeholder

- Hospitals cannot provide intensive post-discharge support to every patient.
- Patients with diabetes may have complex care needs after discharge.
- 30-day readmissions create clinical, operational, and financial burdens.
- The hospital needs a way to prioritize patients for enhanced support.

Stakeholder: Director of Population Health/ Director of Care Management

Decision being supported: Which patients should receive enhanced post-discharge care-management services?

Project Question and Business Value

Net Financial Value = Estimated Savings from Avoided Readmissions - Intervention Cost

- Model identifies high-risk patients.
- Threshold determines how many patients are flagged.
- Cost analysis estimated whether intervention is worth it.
- RAG provides evidence-based discharge and care-transition guidance.

FACE Framework

FACE Component	Application to Project
FRAME	Prioritize diabetes patients for post-discharge support
Analyze	Build readmissions model, tune thresholds, estimate cost/value, develop RAG
Communicate	Present model outputs, cost tradeoffs, and guidance in business-friendly form
Execute/Act	Use the tool to support discharge planning and care management resource allocation

DELTA/DELTA Plus Readiness Assessment

DELTA Element	Application to Project
Data	UCI Diabetes structured encounter data plus CMS/AHRQ discharge-planning documents
Enterprise	Fits into hospital discharge planning, population health, and care-management workflows
Leadership	Requires support from care management, quality, finance, IT, and clinical leadership
Targets	Reduce 30-day readmissions and improve post-discharge resource allocation
Analysts	Data analysts, clinical informaticists, care managers, and operations leaders are needed
Technology	R/Python models, Hugging Face model, FAISS vector search, RAG system, Gradio/Streamlit dashboard
Analytical Techniques	Logistic regression, random forest, threshold tuning, cost/value analysis, fine-tuned classifier, and RAG

UCI Diabetes 130-US Hospitals Dataset

- Dataset: Inpatient encounters involving patients with diabetes
- Total encounters: 101,766
- Target variable: 30-day readmission
- Positive class: **readmitted == "<30"**
- Negative class: **readmitted == "NO" or readmitted == ">30"**

Target Distribution:

Outcome	Count	Percent
No 30-day readmission	90,409	88.8%
30-day readmission	11,357	11.2%

Data Cleaning and Feature Engineering

- Converted "?" values to missing values
- Removed **weight** due to extremely high missingness
- Removed identifier variables:
 - **encounter_id**
 - **patient_nbr**
- Created binary target variable: **readmit30**
- Replaced missing categorical values with "**Unknown**"
- Grouped diagnosis codes into broader categories

Diagnosis groups included:

- Diabetes
- Circulatory
- Respiratory
- Digestive
- Genitourinary
- Musculoskeletal
- Injury
- External/Supplementary
- Other

Model Development and Evaluation

Models:

1. **Logistic Regression**
 - Interpretable baseline
 - Easier to explain
2. **Random Forest**
 - Captures nonlinear patterns and interactions
 - Better risk ranking
3. **Fine-Tuned DistilBERT Classifier**
 - Required fine-tuned model
 - Classifies 30-day readmission from patient text summaries

Evaluation metrics:

- Accuracy
- ROC-AUC
- PR-AUC
- Recall
- Precision
- Confusion matrix
- Cost-based threshold analysis

Key point: The default 0.50 threshold performed poorly because very few patients were predicted as readmitted.

Model Comparison at Best Financial Threshold

Model	Threshold	Recall	Precision	Flagged Patients	True Positives	Net Financial Value
Logistic Regression	0.10	65.1%	15.2%	9,747	1,481	\$1,605,000
Random Forest	0.10	72.0%	14.7%	11,159	1,636	\$1,694,600

***Random Forest was selected because it identified more true readmissions, achieved higher recall, and produced higher estimated net financial value**

Cost-Value: Choosing the Operating Threshold

Assumptions used:

- Cost per readmission: \$12,000
- Cost per intervention: \$200
- Intervention effectiveness: 20%

Net Value = (True Positives X Effectiveness x Readmission Cost) - (Flagged Patients x Intervention Cost)

Random Forest at threshold 0.10:

- Flagged patients: 11,159
- True readmissions identified: 1,636
- Expected readmissions avoided: 327.2
- Estimated savings: \$3,926,400
- Intervention cost: \$2,231,800
- Net financial value: \$1,694,600

*These assumptions should be adjustable in the dashboard

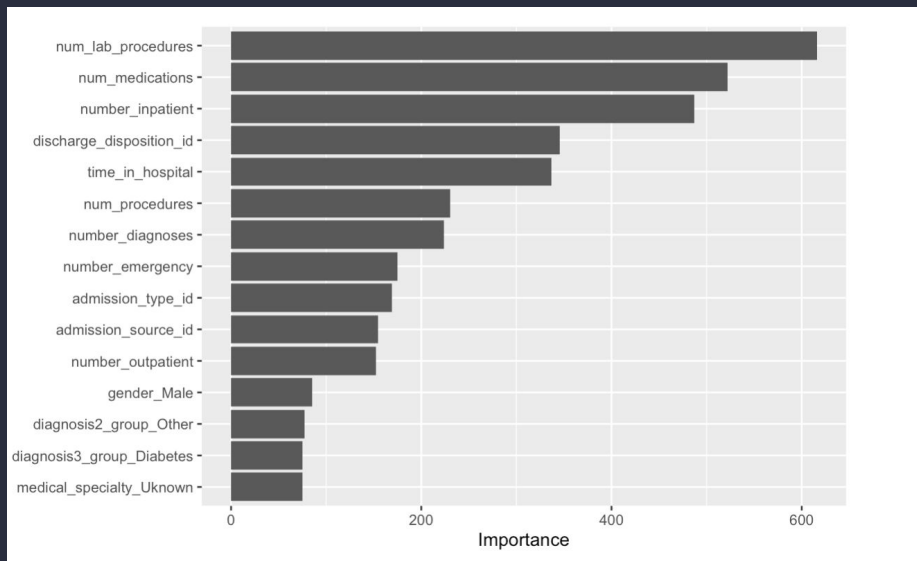
Important Predictors of Readmission Risk

Top predictors included:

- Num_lab_procedures
- Num_medications
- Number_inpatient
- Discharge_disposition_id
- Time_in_hospital
- Num_procedures
- Number_diagnoses
- Number_emergency
- Admission_type_id
- Admission_source_id

Interpretation:

- Clinical complexity matters
- Prior healthcare utilization matters
- Discharge and admission characteristics matter



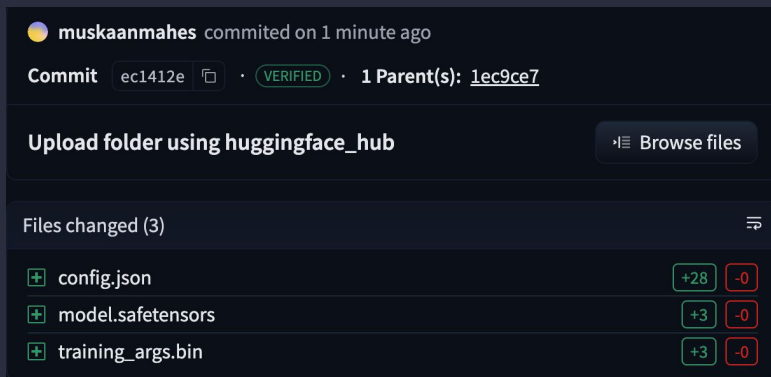
Important caution: Variable importance shows predictive contribution, not causation

Fine-Tuned Readmission Classification Model

Purpose: To satisfy the fine-tuned model requirement and compare a transformer-based classifier with traditional machine learning models

Approach:

- Converted each structured diabetes encounter into a short text summary
- Fine-tuned a DistilBERT text-classification model
- Target variable: **30-day readmission**, Yes vs. No
- Saved the fine-tuned model to Hugging Face for reuse/deployment
- Repository: muskaanmahes/diabetes-readmission-finetuned-classifier
- Uploaded files included **config.json**, **model.safetensors**, and **training_args.bin**



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Upload folder using huggingface_hub Browse files

Files changed (3)

 <code>config.json</code>	+28 -0
 <code>model.safetensors</code>	+3 -0
 <code>training_args.bin</code>	+3 -0

RAG Component: Source-Grounded Discharge Guidance

Purpose: Help care managers connect model-identified risk to evidence-informed discharge and care-transition actions

Documents used:

- CMS Hospital Readmissions Reduction Program materials
- AHRQ RED Toolkit/ discharge-planning resources
- AHRQ care-transition resources

RAG pipeline:

1. Collected CMS/AHRQ documents
2. Loaded CMS webpages and uploaded AHRQ HTML files
3. Cleaned and preprocessed text
4. Split documents into overlapping chunks
5. Created embeddings with sentence transformers
6. Stored chunks in FAISS vector index
7. Retrieved relevant chunks based on stakeholder question
8. Generated source-grounded response

Example Question: What should care managers do for high-risk patients after discharge

RAG Evaluation

Evaluation questions included:

- What discharge planning actions are recommended for high-risk patients?
- What follow-up steps should care managers consider after discharge?
- How can medication review support safer care transitions?
- Why are 30-day readmissions important financially?
- What should a care-management director consider when prioritizing limited resources?

Evaluation criteria:

- Retrieval relevance
- Answer grounded in sources
- Usefulness for business decision
- Limitations or failure modes

Key findings:

- RAG was useful for general discharge-planning and care transition guidance
- Some answers were broad rather than patient-specific
- Source documents support decision-making but do not replace clinical judgment

Deployed Dashboard/Interactive Application

Platform: Hugging Face Spaces

Framework: Gradio/Streamlit

- Loads saved model and generates 30-day readmission-risk predictions
- Allows users to adjust threshold, intervention cost, readmission cost, and intervention effectiveness
- Estimates patients flagged, avoided readmissions, savings, intervention cost, and net financial value
- Includes RAG-based CMS/AHRQ discharge-planning guidance
- Supports care-management decisions about which patients should receive enhanced post-discharge services

Workflow:

Patient data → Risk prediction → Threshold/cost adjustment → Net value estimate → RAG guidance → Care-management action

Deployed Link:

<https://huggingface.co/muskaanmahes/diabetes-readmission-finetuned-classifier/commit/ec1412e90911c2d0a0e69274cacf16a186acaf72#d2h-835064>

Final Recommendation and Limitations

Main recommendation: Use the random forest model with a 0.10 threshold when the goal is to maximize estimated net financial value under the current assumptions

Business recommendation:

- Prioritize patients with predicted readmission risk $\geq 10\%$
- Use enhanced post-discharge services for flagged patients
- Include follow-up calls, medication review, appointment scheduling, and care coordination
- Allow managers to adjust threshold based on staffing capacity
- Use RAG to retrieve discharge-planning guidance for action planning

Limitations:

- Dataset is historical: 1999-2008
- Some variables may not reflect current healthcare workflows
- Readmission prediction is imbalanced and difficult
- Cost assumptions are simplified
- Model does not prove causation
- RAG guidance is broad and source-dependent



Any questions?

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Thank you